



Bayesian stochastic imputation of discontinuous paleoenvironmental proxy records: a case study from Lake Ohrid stable isotopes

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Abstract

Reconstructing past climate variability from lake sediments can be hindered by gaps in multiproxy records. We present a modular Bayesian two-stage approach to impute missing values in a target proxy by leveraging information from all coeval proxies while explicitly propagating uncertainty. As a case study, we reconstruct $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ over the last 624 ka (kiloannum, meaning thousand years) of the Lake Ohrid DEEP sequence, where endogenic-calcite scarcity primarily during glacials leaves substantial isotope gaps. The workflow comprises (i) a latent Gaussian state-space model that maps irregularly sampled proxies onto a common, evenly spaced depth grid (0.25 m in this study) and accommodates reported measurement uncertainties, with optional propagation of any reported uncertainty; and (ii) a multivariate Bayesian linear regression with stochastic search variable selection (SSVS), based on the assumption that each proxy serves as an indirect indicator of multiple environmental processes that identifies a sparse, predictive subset of proxies to reconstruct the isotope series. The reconstructed stable isotope reconstructions exhibit coherent glacial–interglacial variability and gap-bridging skill across data-free intervals, with agreement in adjacent observed segments. By integrating diverse datasets within a coherent probabilistic framework and providing credible intervals for imputed values, the method improves the temporal continuity of paleoenvironmental datasets and offers a promising framework for future studies.

Keywords Bayesian inference · MCMC · Multiproxy data · Paleoclimate · State-space model · Stochastic search variable selection · Time series · Total evidence

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1 Introduction

The reconstruction of past environments relies on the collection and analysis of sedimentary archives; however, drilling projects are time-intensive and costly. Moreover, crucial paleoenvironmental datasets may remain incomplete due to the nature of the sediment or measurement limitations. In this study, we present a modular, Bayesian two-stage approach that leverages all available proxy data—optionally, together with their reported uncertainties—to impute missing values in discontinuous records at a given site. Using the long and continuous sediment sequence from Lake Ohrid on the Balkan Peninsula as a case study, we demonstrate that our method improves the temporal continuity of paleoenvironmental reconstructions, enabling more comprehensive analyses of long-term environmental change. In addition, for applications that require evenly spaced inputs, the framework maps irregular observations onto a common measurement grid, yielding smoothed, commensurate series across proxies.

Lake Ohrid, situated on the Balkan Peninsula, holds the longest continuous drilled sediment record in the eastern Mediterranean, spanning more than one million years (Wagner et al. 2019). As part of an extensive multinational effort, a sediment sequence (DEEP) was recovered within the scope of the International Continental Scientific Drilling Program (ICDP) project, conducted between 2009 and 2013 following several years of hydro-acoustic surveys to identify optimal drilling sites (Wagner et al. 2014). This sequence has been widely studied, yielding a wealth of proxy data that inform past climate and environmental conditions (SCOPSCO 2025).

However, despite the overall continuity of most proxy records, the stable isotope series from the DEEP sequence ($\delta^{18}\text{O}$ and $\delta^{13}\text{C}$) contain temporal gaps where carbonate is scarce in the sediment column (Lacey et al. 2016). Because stable isotopes are mechanistically linked to hydroclimate, they are directly comparable across archives such as speleothems, ice cores, and other lakes (Talbot 1990; Gat 1996; McDermott 2004). Moreover, isotope records can be used in quantitative frameworks, including data assimilation with isotope-enabled models (e.g., Steiger et al. 2017; Dee et al. 2018) and formal meta-analyses (e.g., Zachos et al. 2001; Roberts et al. 2008, 2011). A continuous isotope series therefore enables robust phase/lead-lag analyses, tests of long-distance climatic connections (teleconnections), and integration into isotope-enabled analyses that are generally more straightforward with isotopes than with many other proxy types.

To impute temporal gaps in the Lake Ohrid isotope series, we assume covariate stability (i.e., that relationships between the isotope measurements and the other proxy measurements do not drift substantially at multi-millennial timescales), so that the continuous proxy sets from Lake Ohrid (Section 3.1) and the discontinuous isotope records share a common climate-driven component, with the isotope series chiefly interrupted during glacial intervals over the last 624 ka. Because proxies measured at a single site reflect overlapping aspects of environmental variability, missing values in the stable isotope series ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$) can, in principle, be inferred from their co-variability with the remaining proxies. Building on this premise, we develop a modular Bayesian two-stage framework that treats site-specific proxies as indirect, noisy indicators of underlying environmental processes and propagates uncertainty throughout the analysis.

Because the proxy series have different resolutions and are irregularly sampled in depth, we fit a latent Gaussian state-space model separately for each proxy at its observed depths. Reported measurement uncertainties enter the observation model; otherwise an observation-error variance is estimated. A smooth evolution prior on the latent process provides regularization. We then predict each latent process on a common regular depth grid, retaining posterior means and 95% credible bands.

Using these depth-aligned latent series as predictors, we model the isotope series as responses in a regression with SSVS (O'Hara and Sillanpää 2009; George and McCulloch 1993, 1997). Spike-and-slab priors introduce binary inclusion indicators that shrink uninformative coefficients toward zero, limiting overfitting. Uncertainty from the first stage is propagated by repeating the regression over multiple posterior draws of the latent predictors and combining results. The posterior predictive distribution then yields imputed isotope values with credible intervals.

Our overall explicit and reproducible approach enables the incorporation of all relevant total evidence available (Carnap 1962) for the study site. Moreover, even when using uninformative priors on the regression coefficients, the SSVS framework effectively mitigates concerns arising from the principle of insufficient reason, as discussed by Keynes (1921), by introducing a data-driven mechanism for variable inclusion that avoids assigning equal weight to all potential predictors and instead stochastically evaluates their explanatory relevance to the response variable.

2 Regional setting

Lake Ohrid (Figure 1) is located within an active tectonic graben on the Balkan Peninsula at an altitude of 693 m. It has a maximum depth of 293 m and a total water volume of 55.4 km³ (Wagner et al. 2019; Leng et al. 2010; Francke et al. 2016; Lacey et al. 2016). Approximately 55% of the lake's inflow is derived from karstic springs, with up to 50% of this water originating from the neighboring Lake Prespa to the east. The remaining inflow is supplied by direct precipitation and surface runoff from the surrounding drainage basin (Matzinger et al. 2007).

The region's climate is classified as sub-Mediterranean; however, due to the surrounding mountain ranges, it also exhibits characteristics of a continental climate (Leng et al. 2010; Panagiotopoulos et al. 2013). The area receives approximately 750 mm of precipitation annually, primarily during the winter months, while annual open-water evaporation is around 980 mm (Hollis and Stevenson 1997).

Lake Ohrid is hydrologically open, with outflow occurring primarily through the Black Drin River in the north, which accounts for approximately 60% of total water loss; the remainder is lost through evaporation (Matzinger et al. 2006).

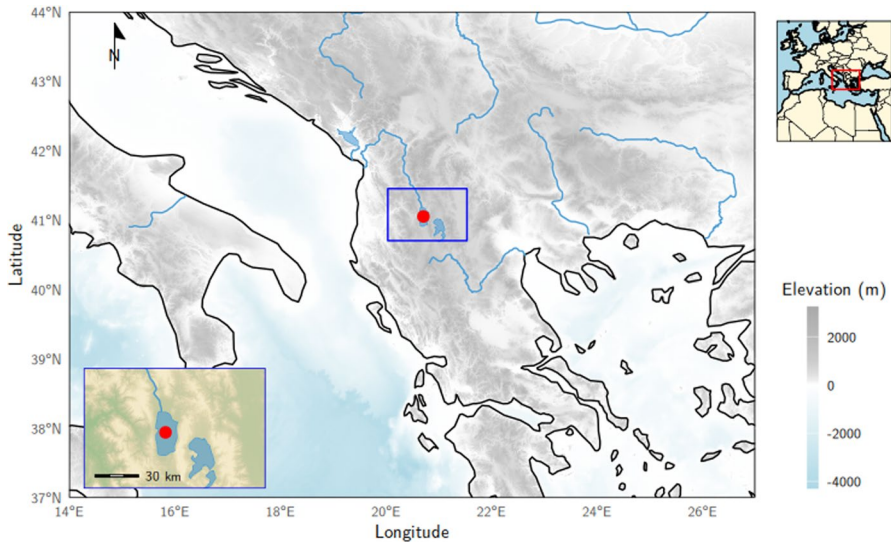


Fig. 1 Location map of Lake Ohrid, with the red dot indicating the DEEP site

3 Materials and methods

3.1 Data

The sediment sequence from the Ohrid DEEP site spans the last 1.35 Ma (Wagner et al. 2019). However, for the purpose of taking advantage of the data published by Francke et al. (2016), we focus specifically on the most recent 624 ka of the record. This dataset includes measurements of magnetic susceptibility (MS), arboreal pollen percentages excluding *Pinus* pollen (AP), total inorganic carbon (TIC), total organic carbon (TOC), total nitrogen (TN), total sulfur (TS), quartz peak area (Qz), biogenic silica (bSiO₂), clay percentage, element counts obtained from X-ray fluorescence (XRF) scanning (Fe, Zr, K, Ca), and stable isotopes of oxygen and carbon ($\delta^{18}\text{O}$ and $\delta^{13}\text{C}$) (Wagner et al. 2019; Francke et al. 2016; Cvetkoska et al. 2021; Just et al. 2019; Lacey et al. 2016; Sadori et al. 2016).

All variables (Figure 2), except for $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$, form uninterrupted records over the studied time interval. However, their measurements differ in resolution and exhibit irregular sampling spacing. In contrast, stable isotope data are available only for limited intervals due to the discontinuous presence of calcite in the sediment record (Lacey et al. 2016). Since calcite primarily occurs during interglacial periods, isotope measurements are available—though not continuously—during parts of MIS 15, MIS 14, MIS 13, MIS 11, MIS 9, MIS 7, and MIS 5¹ (Figure 2). Additionally, an approximately 6 ka-long measurement series is present in MIS-8, spanning

¹ Marine Isotope Stages (MIS) are numbered intervals in marine oxygen-isotope stratigraphy that denote relatively warm (odd-numbered, interglacial) and cold (even-numbered, glacial) periods (Halverson 2013). A commonly noted exception is MIS 3, a relatively warm interstadial within the last glacial cycle.

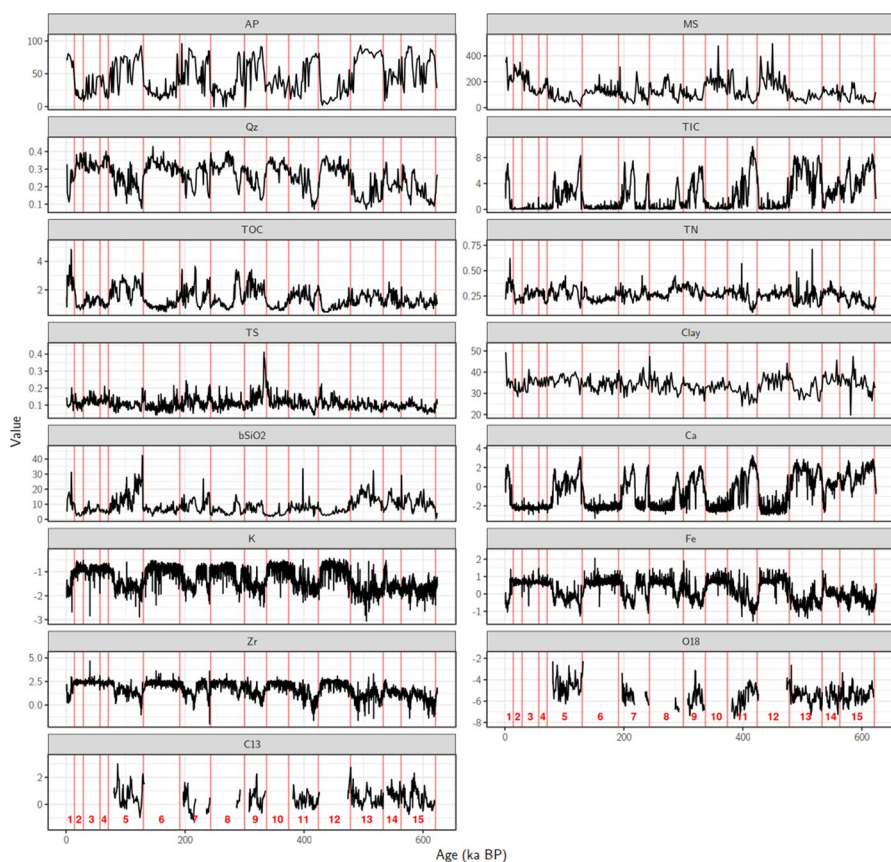


Fig. 2 Raw proxy data from Lake Ohrid spanning the last 624 ka. As described in the text, only the XRF element count data (Ca, K, Fe, and Zr) have been transformed using the centered log-ratio method (Figure S2). Vertical red lines indicate boundaries between Marine isotope stage (MIS) stages, numbered in red

242–236 ka time interval that corresponds to 106–103 m in depth. The intervals without stable isotope measurements are listed in the first column of Table 1.

The published age-depth model for the DEEP site sediment succession was developed in two stages (Wagner et al. 2019; Francke et al. 2016). First, tephrochronological data and tie points were established through cross-correlation between TOC measurements and local summer insolation as well as winter-season length. Then, a Bayesian modeling approach (Blaauw and Christen 2011) was applied, yielding a median age-depth model along with 95% credible intervals, as reported by Wagner et al. (2019).

Given minor discrepancies between the age-depth models presented by Francke et al. (2016) and Wagner et al. (2019), we adjusted the age model of the dataset from Francke et al. (2016) to align precisely with the final model provided by Wagner et al. (2019), using the corresponding depths from both studies.

Table 1 Depth intervals and corresponding temporal lengths used for imputation and model validation. The oxygen and carbon isotope data ($\delta^{18}\text{O}$ and $\delta^{13}\text{C}$) contain six discontinuous intervals requiring imputation

Imputed interval (m)	Temporal length of imputed interval (ka)	Interval used to estimate regression coefficients (m)	Temporal length of training interval (ka)	Interval used for comparison (ka)
32–0.5	80	49.5–32	53	—
96.5–49.5	65	49.5–32	53	218–196
134–87.5	90	146–134	29	218–196
159–134	47	174.5–159	46	335–307
188.5–159.5	47	207.5–188.5	61	427–381
210–190.5	7	243–210	86	533–478

The first column indicates the start and end depths of these intervals, while the second shows their temporal lengths. The third column specifies depth intervals with available isotope data used to estimate regression coefficients (training intervals) for imputing missing data, and the fourth column provides the temporal length of these training intervals. Where possible, regression results were extended into intervals with existing data to evaluate the forecasting performance of the model, as described in Section 3.6. The fifth column lists temporal intervals used for comparison and model validation

Moreover, the original XRF-Zr dataset from Francke et al. (2016) was published as Zr/K ratios. We reconstructed the actual XRF-Zr values by multiplying these ratios with the corresponding XRF-K measurements. It should be noted that the XRF data from Francke et al. (2016) were smoothed using a fifth-order low-pass filter with a cutoff frequency of 0.64, whereas the data reported by Wagner et al. (2019) were smoothed using an 11-point moving average (see Figure S1 for comparison). Despite these methodological differences, both datasets were integrated within the same analytical model. Finally, we applied a centered log-ratio (clr) transformation to the XRF data (Figure S2), under the assumption that they represent compositional data constrained to a sub-simplex (Aitchison 1986).

3.2 Bayesian state-space model for individual proxy data

We model each proxy as a latent Gaussian state-space process observed at irregular depths. To obtain a tractable representation, we introduce a regular depth grid $\{d_t\}_{t=1}^N$ and let the latent states $\{x_t\}$ evolve on this grid. Each observation at depth d_i^{obs} (or, if a time grid is used, at time t_i^{obs}) is mapped to the regular grid via an index $k(i) \in \{1, \dots, N\}$ using the floor function, so that multiple observations may inform the same latent state and some grid cells may remain empty.

Let y_i denote the i th measurement ($i = 1, \dots, N_{\text{obs}}$). We link observations to latent states as

$$y_i \sim N(x_{k(i)}, \sigma_{\text{obs}}^2), \quad i = 1, \dots, N_{\text{obs}}, \quad (1)$$

where N_{obs} is the number of measurements in the proxy series, $k(i)$ maps each observation to its grid cell, and σ_{obs} is the observation standard deviation.

The latent process follows a first-order Gaussian random walk on the regular grid:

$$x_1 \sim N(0, 4^2), \quad (2)$$

$$x_t \sim N(x_{t-1}, \sigma_{\text{proc}}^2), \quad t = 2, \dots, N, \quad (3)$$

where σ_{proc} controls the smoothness of the state evolution per grid step.

We note that the dataset intended for imputation, $\delta^{18}\text{O}$ or $\delta^{13}\text{C}$ in our case, is measured over a shorter time interval than the rest of the proxy data. As a result, the evenly spaced time series inferred from the state-space model for this target variable is temporally shorter than those of the other proxies, which serve as covariates in the regression model described in Section 3.3. While the state-space modeling procedure is applied identically to all variables, we denote the shorter and longer time indices as t_s and t_l , respectively, where $1 < t_s < t_l$, as introduced in Section 3.3.

We use a hierarchical shrinkage prior on the variance components. On the standardized scale, σ_{hyper} and σ_{global} receive half-Student- t ($\nu=1$, i.e., half-Cauchy) hyperpriors with scales 0.2 and 0.5:

$$\sigma_{\text{hyper}} \sim t_1^+(0, 0.2), \quad (4)$$

$$\sigma_{\text{global}} \sim t_1^+(0, 0.5), \quad (5)$$

Conditional on these, the process and observation standard deviations are shrunk toward σ_{global} via truncated normals:

$$\sigma_{\text{proc}} \sim N^+(\sigma_{\text{global}}, \sigma_{\text{hyper}}^2), \quad (6)$$

$$\sigma_{\text{obs}} \sim N^+(\sigma_{\text{global}}, \sigma_{\text{hyper}}^2), \quad (7)$$

where $t_\nu^+(0, s)$ denotes a Student- $t(\nu, 0, s)$ distribution truncated to $[0, \infty)$ and $N^+(\mu, \tau^2)$ denotes a normal distribution with mean μ and variance τ^2 truncated to $[0, \infty)$. This hierarchy provides weakly informative, heavy-tailed regularization while encouraging parsimony via shared shrinkage toward σ_{global} , as advocated by Gelman (2006).

Lastly, to enhance convergence and improve computational efficiency, the latent states were initially standardized. Following model estimation, they were back-transformed to the original scale using the mean and standard deviation of the observed data.

This modeling framework offers a robust approach for reconstructing latent state trajectories, while simultaneously quantifying uncertainty in both the process and observation components. It also enables the transformation of gap-free, irregularly spaced data into an evenly spaced format suitable for further statistical analysis.

Furthermore, this framework can be extended to account for both reported measurement error and uncertainty in the sampling coordinate (depth/age) by introducing a latent categorical time index $t_i^* \in \{1, \dots, N\}$ for each observation $i = 1, \dots, N_{\text{obs}}$. The observed proxy value y_i is treated as a noisy realization of the latent state at the (unknown) grid index t_i^* :

$$y_i \sim N\left(x_{t_i}^*, (\sigma_i^{\text{proxy}})^2 + \sigma_{\text{obs}}^2\right), \quad i = 1, \dots, N_{\text{obs}}, \quad (8)$$

where σ_i^{proxy} is the reported (instrument/analytical) standard deviation for observation i and σ_{obs} is additional observation noise on the standardized scale.

To encode depth/age uncertainty, the latent index t_i^* can be assigned a categorical prior whose probabilities are proportional to a normal kernel centered on the reported coordinate t_i with standard deviation σ_i^{age} :

$$\Pr[t_i^* = t] \propto N(t | t_i, (\sigma_i^{\text{age}})^2), \quad (9)$$

where t represents possible time indices within a restricted search window around t_i . The normalized probability mass function for t_i^* is then computed as follows:

$$t_i^* \sim \text{Cat}(\text{prob}_i), \quad (10)$$

where prob_i represents the precomputed probability mass function over the subset of candidate time indices. This formulation allows the model to account for all types of reported uncertainties while maintaining computational feasibility.

3.3 Regression with SSVS

We regress the target isotope series on the state-space summaries from Section 3.2, i.e., the posterior medians $\hat{x}_t$ (with uncertainties s_t) of the latent states for each proxy on the common grid.

The regression is formulated as a multivariate linear model. In the first part of the model, where the response variable is observed, the regression coefficients are estimated. In the subsequent interval, where the response variable is missing, its values are extrapolated using the available covariates. Let $t = 1, \dots, t_s, \dots, t_l$ denote the evenly spaced domain, where t_s is the final point with observed response data, and t_l is the full extent of the covariate series.

The response variable Y_t , available only over the shorter interval $[1, t_s]$, is represented by the posterior median of the latent states estimated via the state-space model for the proxy to be imputed, $\delta^{18}\text{O}$ or $\delta^{13}\text{C}$ in our case, at depth t , as obtained from the Markov chain Monte Carlo MCMC output. The associated uncertainty Y_t^{sd} corresponds to the posterior standard deviation of the same latent state estimates. The model assumes the following:

$$Y_t \sim N\left(\mu_t, (Y_t^{\text{sd}} + \sigma_y)^2\right), \quad \text{where } t = 1, \dots, t_s. \quad (11)$$

Here, $\mu_t = X_t\beta$ is the expected value of Y_t , calculated as the inner product of the covariate vector X_t and the regression coefficients β .

The covariates X_t , which span the full interval $[1, t_l]$, are similarly derived as the posterior medians of the latent states of other proxy variables estimated by the

state-space model. Their associated uncertainties $X_{t,j}^{\text{sd}}$ are the corresponding posterior standard deviations. Each covariate $X_{t,j}$, indexed by j , is modeled as follows:

$$X_{t,j} \sim N \left(\text{state}_{t,j}, \left(X_{t,j}^{\text{sd}} + \sigma_x \right)^2 \right), \quad (12)$$

where $\text{state}_{t,j}$ denotes the latent true value of the covariate. These latent states are assigned weakly informative Gaussian priors:

$$\text{state}_{t,j} \sim N(0, 100). \quad (13)$$

The prediction process is almost same to the regression for the known response variable, but the only difference is for the unobserved response variable, which is

$$Y_t \sim N \left(\mu_t, (\sigma_y)^2 \right), \quad \text{where } t = t_{s+1}, \dots, t_l, \quad (14)$$

where the standard deviation parameter becomes σ_y , since there is no information coming from the state-space model process.

To perform variable selection, the regression coefficients (β) were decomposed into two components:

$$\beta_j = \gamma_j \delta_j, \quad (15)$$

where $\gamma_j \sim \text{Bernoulli}(\text{prob}_j)$ is a binary inclusion indicator, and δ_j represents the effect size. The inclusion probabilities (prob_j) were assigned uninformative Beta priors:

$$\text{prob}_j \sim \text{Beta}(1, 1). \quad (16)$$

We place a student- t slab on each δ_j to combine aggressive shrinkage of near-zero coefficients with enough tail weight to let truly large effects escape undue penalization:

$$\delta_j \sim t_{\nu_j} \left(0, \sigma_{\text{ss}}^2 \right), \quad (17)$$

where the degrees of freedom

$$\nu_j \sim \text{Exp}(1/30), \quad (18)$$

has mean 30, enabling the model to learn exactly how heavy the tails must be. Small ν_j produce very heavy tails for robustness to outliers or strong signals, while large ν_j recover Gaussian-like behavior near zero (Kruschke 2015). We assign half-Cauchy priors to the scale parameters to further regularize the model (Gelman 2006)

$$\sigma_y \sim t_1^+ \left(0, 4^2 \right), \quad (19)$$

$$\sigma_x \sim t_1^+ (0, 4^2), \quad (20)$$

$$\sigma_{ss} \sim t_1^+ (0, 4^2). \quad (21)$$

3.4 Implementation and inference

For each proxy we standardize the series, build a regular depth grid $\{d_t\}_{t=1}^N$ (0.25 m spacing), and map each irregular observation d_i^{obs} to a grid index $k(i)$. The observation model is $y_i^{\text{std}} \sim N(x_{k(i)}^{\text{std}}, \sigma_{\text{obs}}^2)$ and the latent evolution is a Gaussian random walk $x_t^{\text{std}} | x_{t-1}^{\text{std}} \sim N(x_{t-1}^{\text{std}}, \sigma_{\text{proc}}^2)$ with $x_1^{\text{std}} \sim \mathcal{N}(0, 4^2)$ and hierarchical shrinkage priors on σ_{proc} and σ_{obs} (Section 3.2). We fit this linear-Gaussian state-space model in JAGS (Plummer 2003) via `runjags` (Denwood 2016) in R (R Core Team 2024), sampling from the joint posterior of all parameters and latent states $\{x_t\}$. Conditionally Gaussian blocks are updated from their full conditionals via Gibbs; non-conjugate scale parameters are updated using the generic MCMC steps provided by JAGS. Convergence is assessed by standard diagnostic tools; posterior predictive checks are reported in the Supplement. By “MCMC output” we mean the posterior draws $\{x_t^{(m)}\}_{m=1}^M$ of the latent states. For each grid point t , we summarize these by the posterior median $\hat{x}_t$ and standard deviation s_t , then back-transform to the original scale. These summaries $(\hat{x}_t, s_t)$ provide the inputs and associated uncertainties for the regression stage in Section 3.3. For background on linear state-space modeling, see Harvey (1990), Durbin and Koopman (2012), and Shumway and Stoffer (2025).

3.5 Convergence checking

Model convergence was assessed using the potential scale reduction factor (Gelman and Rubin 1992) supplemented by visual diagnostics such as trace plots and posterior density plots from multiple Markov chains (see supplementary figures).

To evaluate the fit of the SSVS model, posterior predictive distributions ($Y_{\text{rep},t}$) were generated for the training interval:

$$Y_{\text{rep},t} \sim N(\mu_t, (Y_t^{\text{sd}} + \sigma_y)^2), \quad \text{where } t = t_1, \dots, t_s. \quad (22)$$

The corresponding posterior predictive check PPC plots are presented in Figures S7, S16, S26, S36, S46, S56, S66, S75, S85, S95, S105, and S115.

3.6 Model assessment

To assess the model’s forecasting performance, we performed predictions in intervals, where observational data are available (Table 1), specifically covering the intervals 96.5–87.5 m (218–196 ka), 146–134 m (335–307 ka), 174.5–159.5 m (427–381 ka), and 207.5–190.5 m (533–478 ka). Additional PPCs were conducted to compare the

predictive densities of the state-space and SSVS models. These PPCs, along with Bayesian p -values and credible interval coverage statistics, offer a comprehensive evaluation of model calibration and predictive accuracy (Gelman et al. 1996, 2013)

The results of these diagnostics, presented in Figures S22, S32, S42, S52, S62, S81, S91, S101, S111, and S121, demonstrate that the Bayesian inference framework applied here is both predictive and well-calibrated, supporting its overall robustness.

4 Results

PPCs and the model's fitted values for the training interval indicate that the model reproduces the training data well (see supplementary figures). Moreover, posterior state-space estimates for intervals with observations—218–196, 335–307, 427–381, and 533–478 ka—closely track the corresponding regression-based predictions, supporting the suitability of the approach for inferring stable -isotope values (Figures S19, S29, S39, S49, S59, S78, S88, S98, S108, and S118).

The complete reconstruction (Figure 3) spans the last 624 ka and encompasses multiple glacial–interglacial cycles. Across the imputed intervals, both $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ exhibit quasi-periodic variability at orbital scales, consistent with well-documented global stacks.

5 Discussion

Paleoenvironmental reconstructions rely on sampling and analyzing geological archives whose measurements are interpreted as proxies for past conditions. These proxies—geochemical, micropaleontological, and geophysical among others (cf. Bradley 2015; Fischer and Wefer 1999; Buczkó et al. 2009)—are indirect indicators that integrate multiple environmental processes (Ön and Özeren 2019). They may be quantitatively calibrated to climate variables (e.g., Bemis et al. 1998; Overpeck et al. 1985) or interpreted jointly in multiproxy frameworks (Birks and Birks 2006).

Even where comprehensive records exist at a site, individual series can contain gaps or exhibit low resolution over specific intervals. Such discontinuities may arise from the absence of suitable material for analysis, the inherent properties of the archive, or the high cost and practical limits of certain analytical techniques.

We propose a proxy-agnostic statistical framework for imputing missing values, provided that (i) the predictor set and the target proxy share a common, climate-driven component at the timescales of interest, and (ii) there is sufficient temporal overlap to learn a stable mapping.

Under these conditions, the method is not tied to any single proxy type. As a demonstration, we apply the method to the Lake Ohrid DEEP sediment sequence, focusing on the last 624 ka to leverage the XRF-derived Zr and K datasets of Francke et al. (2016), which are available only for this interval. Among the reported proxies, the stable isotope ratios of oxygen and carbon ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$) contain gaps owing to the discontinuous presence of endogenic calcite in the sedimentary record (Wagner et al. 2019; Lacey et al. 2016). Our approach treats the remaining proxies as noisy, indirect

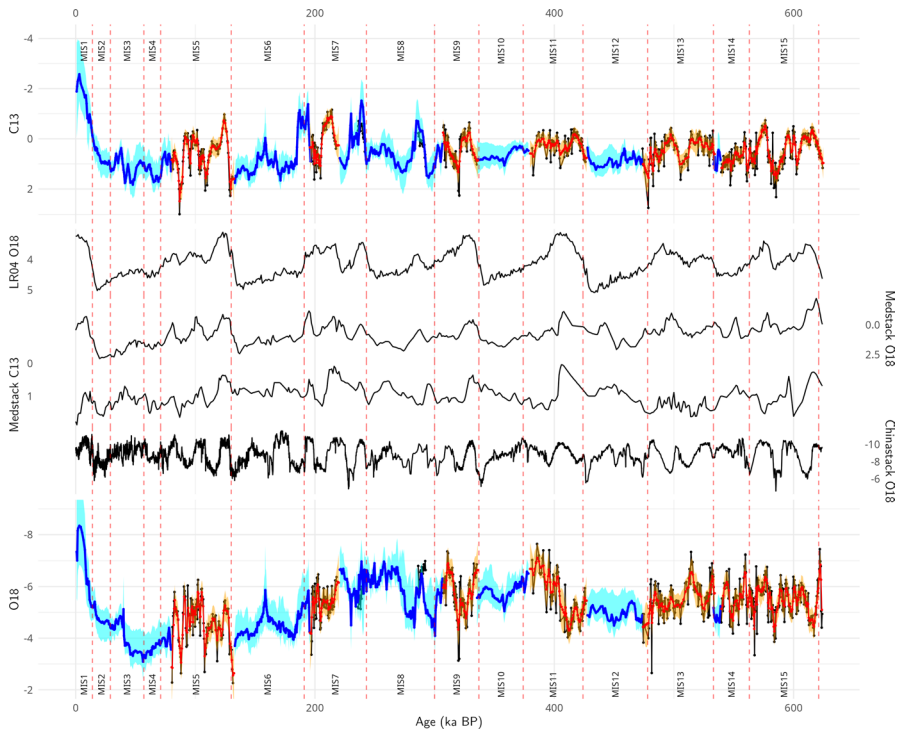


Fig. 3 Comparison of the combined results with global isotope records. From top to bottom: Lake Ohrid $\delta^{13}\text{C}$, LR04 benthic $\delta^{18}\text{O}$ stack (Lisiecki and Raymo 2005), Medstack planktonic $\delta^{18}\text{O}$ (Wang et al. 2010), Medstack planktonic $\delta^{13}\text{C}$ (Wang et al. 2010), Chinese speleothem stack $\delta^{18}\text{O}$ (Cheng et al. 2016), and Lake Ohrid $\delta^{18}\text{O}$. All isotope values are reported in per mille (‰) relative to the Vienna Pee Dee belemnite (VPDB) standard. Blue lines and cyan shading on the Lake Ohrid plots represent imputed intervals obtained through regression, along with 89% credible intervals. Red lines and orange shading indicate the state-space model results and their associated credible intervals

indicators of the underlying environmental drivers and uses them to reconstruct the isotope series probabilistically, while propagating uncertainty.

Although multiple processes contribute, the stable $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ data which are widely used as paleoenvironmental proxies in lacustrine studies, primarily reflect, respectively, the precipitation–evaporation (P/E) balance and in-lake productivity (and, by extension, soil development); both are closely linked to climate variables such as humidity and temperature (Gat 1996; Urey 1948; Dansgaard 1964; Leng and Marshall 2004; Roberts et al. 2008; Gierlowski-Kordesch 2010; Lacey et al. 2015, 2016). Moreover, the additional proxy datasets used here to impute the isotope series are themselves climate-sensitive. While multiple environmental factors contribute to the observed signals, the likely controls on variability at Lake Ohrid and at many other locations have been discussed extensively and can be summarized as follows:

1. TOC, TIC, TN, TS, bSiO₂, and Ca measurements are chiefly controlled by in-lake organic productivity (and associated carbonate production), which responds to environmental conditions such as temperature, runoff, precipitation, and

- nutrient availability (e.g., Vogel et al. 2010; Francke et al. 2016; Wagner et al. 2009; Cvetkoska et al. 2021; Davies et al. 2015; Danladi et al. 2021);
2. K, Fe, Qz peak area, clay, and MS measurements serve primarily as proxies for clastic/terrigenous input and sedimentary facies, while Zr counts often track aeolian influx; these are modulated by catchment runoff and hence by precipitation and drought conditions (e.g., Vogel et al. 2010; Francke et al. 2016; Wagner et al. 2019; Eris et al. 2018; Danladi et al. 2023; Davies et al. 2015);
 3. In intervals with limited or no anthropogenic influence, AP percentage reflects tree cover and serves as an indicator of effective moisture and precipitation (e.g., Sadori et al. 2016; Wagner et al. 2019).

Furthermore, prior work documents broadly continuous relationships between isotope records and other proxies across glacial–interglacial transitions in a range of lake systems (e.g., Francke et al. 2022; Torfstein et al. 2015; Menking et al. 1997; Chaplignin et al. 2012), including Lake Prespa—Ohrid’s neighbor (Leng et al. 2013)—and Lake Ohrid itself (Lacey et al. 2016; Wagner et al. 2019).

Given these studies, we assume that each proxy reflects a mixture of a limited set of latent environmental drivers operating at the multi-millennial timescales of interest. Each series is thus a distinct linear combination (with proxy-specific weights) of the same underlying drivers; although the weights differ, the shared origins imply a structured interdependence. Under this assumption, information can be borrowed across series to reconstruct gaps in a target proxy.

Therefore, due to the mechanistic linkages, and the shared climatic forcing, we leverage covariance among non-isotopic proxies to impute missing isotope values. Concretely, we adopt a modular two-stage Bayesian workflow: a latent Gaussian state-space step maps irregular observations—together with any reported measurement uncertainties—onto a common 0.25 m depth grid via a selection operator, attenuating observation noise and yielding smoothed, commensurate inputs across proxies. Subsequently, to identify the most relevant predictors, we employ a multivariate regression with SSVS, which enables selecting only predictors that improve predictive performance (out-of-sample) for the target isotope series under uncertainty. The SSVS approach further enhances the regression model by incorporating hierarchical priors on the regression coefficients, governed by latent indicator variables that determine the inclusion of each predictor. Predictors that are stochastically irrelevant are effectively shrunk toward zero through small variances, thereby reducing the risk of overfitting and improving model interpretability. This modular Bayesian two-stage framework propagates uncertainty throughout the entire inference process, resulting in reliable and statistically robust imputations. In this way, imputation targets the climate-coherent component shared across proxies while propagating uncertainty via the posterior predictive distribution; the goal is not to replicate high-frequency, process-specific anomalies unique to individual proxies, but to reconstruct the latent climatic signal supported by the multivariate evidence.

Stable isotope data from the Lake Ohrid DEEP sequence were measured only for parts of several interglacial intervals (odd-numbered MIS in the marine isotope stratigraphy), with no measurements available for MIS-1 (Figure 3). Consequently, the imputation presented here primarily fills gaps within glacial intervals. Validation

of the results is non-trivial because observations are largely absent in those intervals, except for short segments in MIS-7 and MIS-8 (Figures S23 and S82). In these short segments, the model reproduces both $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ with good fidelity (Figures S29 and S88).

To further assess predictive skill across missing spans, we performed gap-bridging forecasts: for each target gap, the model was fit using data up to the gap boundary, predictions were propagated across the data-free interval using the continuous proxy predictors, and forecasting continued into the subsequent interval where isotope observations exist (Figures S19, S29, S39, S49, S59, S78, S88, S98, S108, and S118). The predicted series agrees closely with the observed values over tens-of-kiloyear overlaps, indicating that the model recovers low-frequency structure as well as intermediate variability.

We also conducted proxy-omission (leave-one-proxy-out) tests: for each run, we randomly selected one proxy, masked an interval used for isotope imputation, and re-estimated the model using the remaining proxies as predictors. As expected, removing a predictor modestly degrades performance; nevertheless, reconstructions remain acceptable—particularly for bSiO_2 , Qz, TIC, TOC, and TN—and, across all proxies, the imputed series reproduce the variability in the observed segments (Figures S122–S128), supporting robustness to changes in proxy availability.

Overall, the final reconstructions of the isotope datasets capture both glacial–interglacial variability and fluctuations within glacial periods, showing broad agreement with global reference records. Predictions rely on a covariate stability assumption that the conditional isotope–proxy relationship is stable over millennial scales, and are presented with credible intervals to reflect this uncertainty.

Imputation/regularization is routine in Earth sciences, from age-depth models (e.g., Blaauw et al. 2018) and climate reanalyses (e.g., Harris et al. 2020) to historical gridded reconstructions (e.g., Cook et al. 2015) and time-series completion (e.g., Barker et al. 2011). Here, we use Bayesian data augmentation to generate posterior predictions for missing $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ values of Lake Ohrid, with full uncertainty propagation, so that isotope-specific analyses (e.g., cross-archive comparisons and isotope-enabled model-data evaluation) remain possible without discarding Lake Ohrid from those workflows. Because $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ measurements are recorded across many archives and are simulated explicitly by isotope-enabled models, they provide a practical “common currency” for data–data and data–model comparisons, something most other proxies cannot offer without site-specific calibration. Our approach does not add new observations or “extra evidence”; instead, it summarizes what the observed proxies jointly imply about the unobserved isotope values and reports that as uncertainty-aware posterior distributions (cf. Rochefort-Maranda 2016). To prevent pseudo-replication, (i) imputed intervals are clearly flagged and always shown with credible bands; (ii) imputed isotopes are not and should not be used as independent evidence alongside the proxies that informed them.

Because proxy–isotope relations may be non-stationary and process-specific signals are unrecoverable from other series, our imputation explicitly targets only the climate-coherent component shared across proxies. The completed $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ are thus a convenience product for tasks that specifically require isotopes and should be analyzed as ensembles, not as deterministic observations. This modular, uncer-

tainty-propagating baseline preserves comparability in isotope space and paves the way for more precise joint hierarchical models (e.g., with time-varying coefficients) as data allow.

Ultimately, although we demonstrated the methodology on a single sediment core from Lake Ohrid, it is directly transferable to other lacustrine archives that exhibit proxy discontinuities. For example, the Lake Acıgöl (southwest Anatolia) record contains intervals where carbonate microfossils are absent and lithological alternations preclude continuous isotope measurements (Çaldırak et al. 2025; Akçer-Ön et al. 2017). The framework also extends to multi-core applications at a single site. For example, at Lake Bourget (France), gaps in the longer core (LDB04-I) can be constrained by jointly modeling it with the more complete short core (LDB01-I) (Arnaud et al. 2012). More broadly, the model can be applied to networks of the same proxy across sites that share the similar environmental information, provided robust age models allow alignment and age uncertainty is propagated (cf. Ön et al. 2021). Finally, measurement uncertainty can be incorporated via an errors-in-variables formulation, for example, propagating age model uncertainty when harmonizing samples within or across sites, and including instrument-reported analytical errors as in Ön et al. (2025). This yields posterior predictions that explicitly account for these uncertainty sources.

6 Conclusions

We introduced a modular, Bayesian two-stage methodology for imputing missing paleoenvironmental proxy data. In the first stage, each available proxy series is mapped to a common, regular depth grid using a latent Gaussian state-space model, which can incorporate reported measurement uncertainty. In the second stage, a multivariate regression with SSVS imputes missing values in the target series from the temporally complete proxies while propagating uncertainty.

Applied to the Lake Ohrid DEEP sequence, the approach addresses irregular sampling in depth and yields continuous $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ reconstructions that capture glacial–interglacial variability and intra-glacial fluctuations consistent with global reference records. While the method assumes approximately stable proxy–isotope relationships at multi-millennial timescales, it is flexible across proxy types and accommodates reported measurement uncertainties, making it broadly applicable to diverse paleoarchives.

Overall, our approach offers two key advantages.

1. It provides a Bayesian framework for regularizing depth- or age-irregular proxy series, using all available information, including reported uncertainties, thereby producing commensurate inputs for further analyses.
2. It delivers calibrated posterior imputations with credible intervals by integrating total evidence across multiple proxies via SSVS, reducing overfitting through sparse, data-driven predictor selection.

The completed isotope series are posterior predictive ensembles, not deterministic observations, and must not be treated as independent evidence alongside the proxies that informed their imputation.

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Data availability The code and the data from multiple studies can be found at https://github.com/zboraoon/proxy_imputation_SSVS.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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References

- Aitchison J (1986) The statistical analysis of compositional data. Monographs on statistics and applied probability. Chapman & Hall, Boca Raton. <https://doi.org/10.1007/978-94-009-4109-0>
- Akçer-Ön S, Sakiñç M, Helvacı C, et al. (2017) Quaternary paleoclimate of southwest Anatolia and its surroundings: Acigöl (Denizli-Afyonkarahisar) and Lake Salda (Burdur) climate records and comparison with global records (in Turkish). Tech. rep., TÜBİTAK, project no: 113Y408, <https://search.trdizin.gov.tr/en/yayin/detay/616643/guneybati-anadolu-ve-civarinin-kuvaterner-donemi-paleoiklimi-acigol-denizli-afyonkarahisar-ve-salda-golu-nde-burdur-iklim-kayıtlari-ve-kuresel-kayıtlar-ile-karsilastirilmesi>
- Arnand F, Révillon S, Debret M, et al (2012) Lake Bourget regional erosion patterns reconstruction reveals Holocene NW European Alps soil evolution and paleohydrology. Quatern Sci Rev 51:81–92. <https://doi.org/10.1016/j.quascirev.2012.07.025>
- Barker S, Knorr G, Edwards RL et al (2011) 800,000 years of abrupt climate variability. Science 334(6054):347–351. <https://doi.org/10.1126/science.1203580>

- Bemis BE, Spero HJ, Bijma J et al (1998) Reevaluation of the oxygen isotopic composition of planktonic foraminifera: experimental results and revised paleotemperature equations. *Paleoceanography* 13(2):150–160. <https://doi.org/10.1029/98PA00070>
- Birks HH, Birks HJB (2006) Multi-proxy studies in palaeolimnology. *Veg Hist Archaeobotany* 15(4):235–251. <https://doi.org/10.1007/s00334-006-0066-6>
- Blaauw M, Christen JA (2011) Flexible paleoclimate age-depth models using an autoregressive gamma process. *Bayesian Anal* 6(3):457–474. <https://doi.org/10.1214/ba/1339616472>
- Blaauw M, Christen JA, Bennett KD et al (2018) Double the dates and go for Bayes—impacts of model choice, dating density and quality on chronologies. *Quatern Sci Rev* 188:58–66. <https://doi.org/10.1016/j.quascirev.2018.03.032>
- Bradley RS (2015) *Paleoclimatology: reconstructing climates of the quaternary*, 3rd edn. Academic Press, Cambridge
- Buczkó K, Korponai J, Padisák J, et al (eds) (2009) *Palaeolimnological proxies as tools of environmental reconstruction in fresh water*. Springer, Berlin. <https://doi.org/10.1007/978-90-481-3387-1>
- Carnap R (1962) *Logical foundations of probability*, 2nd edn. The University of Chicago Press, Chicago
- Çaldırak H, Ön ZB, Akkiraz MS, et al (2025) Chronology and environmental changes from a sediment core spanning the last 485 ka from Lake Acigöl (SW Anatolia). *Quatern Int* 722:109710. <https://doi.org/10.1016/j.quaint.2025.109710>
- Chapligin B, Meyer H, Swann GEA et al (2012) A 250 ka oxygen isotope record from diatoms at Lake El'gygytgyn, far east Russian Arctic. *Clim Past* 8(5):1621–1636. <https://doi.org/10.5194/cp-8-1621-2012>
- Cheng H, Edwards RL, Sinha A et al (2016) The Asian monsoon over the past 640,000 years and ice age terminations. *Nature* 534(7609):640–646. <https://doi.org/10.1038/nature18591>
- Cook ER, Seager R, Kushnir Y et al (2015) Old World megadroughts and pluvials during the Common Era. *Sci Adv* 1(10):e1500561. <https://doi.org/10.1126/sciadv.1500561>
- Cvetkoska A, Jovanovska E, Hauße T et al (2021) Drivers of phytoplankton community structure change with ecosystem ontogeny during the Quaternary. *Quatern Sci Rev* 265:107046. <https://doi.org/10.1016/j.quascirev.2021.107046>
- Danladi IB, Akçer-Ön S, Ön ZB et al (2021) High-resolution temperature and precipitation variability of southwest Anatolia since 1730 CE from Lake Gölcük sedimentary records. *Turk J Earth Sci* 30(5):601–610. <https://doi.org/10.3906/yer-2008-14>
- Danladi IB, Akçer-Ön S, Litt T et al (2023) A late Holocene climate reconstruction from the high-altitude Lake Gölcük sedimentary records, Isparta (SW Anatolia). *Quatern Res* 115:120–133. <https://doi.org/10.1017/qua.2023.26>
- Dansgaard W (1964) Stable isotopes in precipitation. *Tellus* 16(4):436–468. <https://doi.org/10.1111/j.2153-3490.1964.tb00181.x>
- Davies SJ, Lamb HF, Roberts SJ (2015) Micro-XRF core scanning in palaeolimnology: recent developments. In: Croudace IW, Rothwell RG (eds) *Micro-XRF studies of sediment cores: applications of a non-destructive tool for the environmental sciences*. Springer, Dordrecht, pp 189–226. https://doi.org/10.1007/978-94-017-9849-5_7
- Dee SG, Russell JM, Morrill C et al (2018) PRYSM v2.0: A proxy system model for lacustrine archives. *Paleoceanogr Paleoclimatol* 33(11):1250–1269. <https://doi.org/10.1029/2018PA003413>
- Denwood MJ (2016) runjags: An R package providing interface utilities, model templates, parallel computing methods and additional distributions for MCMC models in JAGS. *J Stat Softw* 71(9):1–25. <https://doi.org/10.18637/jss.v071.i09>
- Durbin J, Koopman SJ (2012) *Time series analysis by state space methods*. Oxford University Press, Oxford. <https://doi.org/10.1093/acprof:oso/9780199641178.001.0001>
- Eris KK, Ön SA, Çagatay MN, et al (2018) Late Pleistocene to Holocene paleoenvironmental evolution of Lake Hazar, Eastern Anatolia, Turkey. *Quatern Int* 486:4–16. <https://doi.org/10.1016/j.quaint.2017.09.027>
- Fischer G, Wefer G (eds) (1999) *Use of proxies in paleoceanography: examples from the South Atlantic*. Springer, Berlin. <https://doi.org/10.1007/978-3-642-58646-0>
- Francke A, Wagner B, Just J et al (2016) Sedimentological processes and environmental variability at Lake Ohrid (Macedonia, Albania) between 637 ka and the present. *Biogeosciences* 13(4):1179–1196. <https://doi.org/10.5194/bg-13-1179-2016>
- Francke A, Lacey JH, Marchegiano M et al (2022) Last Glacial central Mediterranean hydrology inferred from Lake Trasimeno's (Italy) calcium carbonate geochemistry. *Boreas* 51(1):219–237. <https://doi.org/10.1111/bor.12552>

- Gat JR (1996) Oxygen and hydrogen isotopes in the hydrologic cycle. *Annu Rev Earth Planet Sci* 24:225–262. <https://doi.org/10.1146/annurev.earth.24.1.225>
- Gelman A (2006) Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Anal* 1(3):515–534. <https://doi.org/10.1214/06-BA117A>
- Gelman A, Rubin DB (1992) Inference from iterative simulation using multiple sequences. *Stat Sci* 7(4):457–472. <https://doi.org/10.1214/ss/1177011136>
- Gelman A, Meng XL, Stern H (1996) Posterior predictive assessment of model fitness via realized discrepancies. *Stat Sin* 6(4):733–760
- Gelman A, Carlin JB, Stern HS et al (2013) *Bayesian data analysis*, 3rd edn. CRC Press, Boca Raton. <https://doi.org/10.1201/b16018>
- George EI, McCulloch RE (1993) Variable selection via Gibbs sampling. *J Am Stat Assoc* 88(423):881–889. <https://doi.org/10.1080/01621459.1993.10476353>
- George EI, McCulloch RE (1997) Approaches for Bayesian variable selection. *Stat Sin* 7(2):339–373
- Gierlowski-Kordesich EH (2010) Chapter 1 lacustrine carbonates. In: Alonso-Zarza AM, Tanner LH (eds) *Carbonates in continental settings: facies, environments, and processes, developments in sedimentology*, vol 61. Elsevier, Amsterdam, pp 1–101. [https://doi.org/10.1016/S0070-4571\(09\)06101-9](https://doi.org/10.1016/S0070-4571(09)06101-9)
- Halverson GP (2013) Marine isotope stratigraphy. In: Rink WJ, Thompson J (eds) *Encyclopedia of scientific dating methods*. Springer, Dordrecht. https://doi.org/10.1007/978-94-007-6304-3_130
- Harris I, Osborn TJ, Jones P et al (2020) Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. *Sci Data*. <https://doi.org/10.1038/s41597-020-0453-3>
- Harvey AC (1990) *Forecasting, structural time series models and the Kalman Filter*. Cambridge University Press, Cambridge
- Hollis GE, Stevenson AC (1997) The physical basis of the Lake Mikri Prespa systems: geology, climate, hydrology and water quality. *Hydrobiologia* 351(1):1–19. <https://doi.org/10.1023/A:1003067115862>
- Just J, Sagnotti L, Nowaczyk NR et al (2019) Recordings of fast paleomagnetic reversals in a 1.2 Ma greigite-rich sediment archive from Lake Ohrid. *Balkans J Geophys Res Solid Earth* 124(12):12445–12464. <https://doi.org/10.1029/2019JB018297>
- Keynes JM (1921) *A treatise on probability*. Macmillan, New York
- Kruschke JK (2015) *Doing Bayesian data analysis*, 2nd edn. Academic Press, Boston
- Lacey JH, Francke A, Leng MJ et al (2015) A high-resolution late glacial to Holocene record of environmental change in the Mediterranean from Lake Ohrid (Macedonia/Albania). *Int J Earth Sci* 104(6):1623–1638. <https://doi.org/10.1007/s00531-014-1033-6>
- Lacey JH, Leng MJ, Francke A et al (2016) Northern Mediterranean climate since the Middle Pleistocene: a 637 ka stable isotope record from Lake Ohrid (Albania/Macedonia). *Biogeosciences* 13(6):1801–1820. <https://doi.org/10.5194/bg-13-1801-2016>
- Leng MJ, Marshall JD (2004) Palaeoclimate interpretation of stable isotope data from lake sediment archives. *Quatern Sci Rev* 23(7):811–831. <https://doi.org/10.1016/j.quascirev.2003.06.012>
- Leng MJ, Banerjee I, Zanchetta G et al (2010) Late Quaternary palaeoenvironmental reconstruction from Lakes Ohrid and Prespa (Macedonia/Albania border) using stable isotopes. *Biogeosciences* 7(10):3109–3122. <https://doi.org/10.5194/bg-7-3109-2010>
- Leng MJ, Wagner B, Boehm A et al (2013) Understanding past climatic and hydrological variability in the Mediterranean from Lake Prespa sediment isotope and geochemical record over the Last Glacial cycle. *Quatern Sci Rev* 66:123–136. <https://doi.org/10.1016/j.quascirev.2012.07.015>
- Lisiecki LE, Raymo ME (2005) A Pliocene-Pleistocene stack of 57 globally distributed benthic $\delta^{18}\text{O}$ records. *Paleoceanography* 20(1). <https://doi.org/10.1029/2004PA001071>
- Matzinger A, Spirkovski Z, Patceva S et al (2006) Sensitivity of ancient Lake Ohrid to local anthropogenic impacts and global warming. *J Great Lakes Res* 32(1):158–179. [https://doi.org/10.3394/0380-1330\(2006\)32\[158:SOALOT\]2.0.CO;2](https://doi.org/10.3394/0380-1330(2006)32[158:SOALOT]2.0.CO;2)
- Matzinger A, Schmid M, Veljanoska-Sarafiloska E et al (2007) Eutrophication of ancient Lake Ohrid: global warming amplifies detrimental effects of increased nutrient inputs. *Limnol Oceanogr* 52(1):338–353. <https://doi.org/10.4319/lo.2007.52.1.0338>
- McDermott F (2004) Palaeo-climate reconstruction from stable isotope variations in speleothems: a review. *Quatern Sci Rev* 23(7):901–918. <https://doi.org/10.1016/j.quascirev.2003.06.021>
- Menking KM, Bischoff JL, Fitzpatrick JA et al (1997) Climatic/Hydrologic oscillations since 155,000 yr B.P. at Owens Lake, California, reflected in abundance and stable isotope composition of sediment carbonate. *Quatern Res* 48(1):58–68. <https://doi.org/10.1006/qres.1997.1898>
- O'Hara RB, Sillanpää MJ (2009) A review of Bayesian variable selection methods: what, how and which. *Bayesian Anal* 4(1):85–117. <https://doi.org/10.1214/09-BA403>

- Ön ZB, Özeren MS, (2019) Temperature and precipitation variability in eastern Anatolia: results from independent component analysis of Lake Van sediment data spanning the last 250 kyr BP. *Quatern Int* 514:119–129. <https://doi.org/10.1016/j.quaint.2018.11.037>
- Ön ZB, Ateş ME, Kaiser J (2025) A conceptual and statistical framework for delineating the timing of a stratigraphic transition: Holocene-Anthropocene boundary as a case study. *Progress Phys Geogr Earth Environ* 49(1–2):63–83. <https://doi.org/10.1177/03091333241310742>
- Ön ZB, Greaves AM, Akçer-Ön S et al (2021) A Bayesian test for the 4.2 ka BP abrupt climatic change event in southeast Europe and southwest Asia using structural time series analysis of paleoclimate data. *Clim Change* 165(7):1–19. <https://doi.org/10.1007/s10584-021-03010-6>
- Overpeck JT, Webb T, Prentice IC (1985) Quantitative Interpretation of Fossil Pollen Spectra: Dissimilarity Coefficients and the Method of Modern Analogs. *Quatern Res* 23(1):87–108. [https://doi.org/10.1016/0033-5894\(85\)90074-2](https://doi.org/10.1016/0033-5894(85)90074-2)
- Panagiotopoulos K, Aufgebauer A, Schäbitz F, et al (2013) Vegetation and climate history of the Lake Prespa region since the Lateglacial. *Quatern Int* 293:157–169. <https://doi.org/10.1016/j.quaint.2012.05.048>
- Plummer M (2003). JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. Vienna, Austria
- R Core Team (2024) R: a language and environment for statistical computing. R Foundation for Statistical Computing. Vienna, Austria. www.R-project.org
- Roberts N, Jones MD, Benkaddour A et al (2008) Stable isotope records of late Quaternary climate and hydrology from Mediterranean lakes: the ISOMED synthesis. *Quatern Sci Rev* 27(25):2426–2441. <https://doi.org/10.1016/j.quascirev.2008.09.005>
- Roberts N, Eastwood WJ, Kuzucuoglu C et al (2011) Climatic, vegetation and cultural change in the eastern Mediterranean during the mid-Holocene environmental transition. *The Holocene* 21(1):147–162. <https://doi.org/10.1177/0959683610386819>
- Rocheft-Maranda G (2016) How we load our data sets with theories and why we do so purposefully. *Stud Hist Philos Sci Part A* 60:1–6. <https://doi.org/10.1016/j.shpsa.2016.08.002>
- Sadori L, Koutsodendris A, Panagiotopoulos K et al (2016) Pollen-based paleoenvironmental and paleoclimatic change at Lake Ohrid (south-eastern Europe) during the past 500 ka. *Biogeosciences* 13(5):1423–1437. <https://doi.org/10.5194/bg-13-1423-2016>
- SCOPSCO (2025) Publications. <https://www.icdp-online.org/projects/by-continent/europe/scopco-macedonia/publications>
- Shumway RH, Stoffer DS (2025) Time series analysis and its applications: with R examples, 5th edn. Springer, Berlin
- Steiger NJ, Steig EJ, Dee SG et al (2017) Climate reconstruction using data assimilation of water isotope ratios from ice cores. *J Geophys Res Atmos* 122(3):1545–1568. <https://doi.org/10.1002/2016JD026011>
- Talbot MR (1990) A review of the palaeohydrological interpretation of carbon and oxygen isotopic ratios in primary lacustrine carbonates. *Chem Geol Isotope Geosci Sect* 80(4):261–279. [https://doi.org/10.1016/0168-9622\(90\)90009-2](https://doi.org/10.1016/0168-9622(90)90009-2)
- Torfstein A, Goldstein SL, Kushnir Y et al (2015) Dead Sea drawdown and monsoonal impacts in the Levant during the last interglacial. *Earth Planet Sci Lett* 412:235–244. <https://doi.org/10.1016/j.epsl.2014.12.013>
- Urey HC (1948) Oxygen isotopes in nature and in the laboratory. *Science* 108(2810):489–496. <https://doi.org/10.1126/science.108.2810.489>
- Vogel H, Wagner B, Zanchetta G et al (2010) A paleoclimate record with tephrochronological age control for the last glacial-interglacial cycle from Lake Ohrid, Albania and Macedonia. *J Paleolimnol* 44(1):295–310. <https://doi.org/10.1007/s10933-009-9404-x>
- Wagner B, Lotter AF, Nowaczyk N et al (2009) A 40,000-year record of environmental change from ancient Lake Ohrid (Albania and Macedonia). *J Paleolimnol* 41(3):407–430. <https://doi.org/10.1007/s10933-008-9234-2>
- Wagner B, Wilke T, Krastel S et al (2014) The SCOPSCO drilling project recovers more than 1.2 million years of history from Lake Ohrid. *Sci Drill* 17:19–29. <https://doi.org/10.5194/sd-17-19-2014>
- Wagner B, Vogel H, Francke A et al (2019) Mediterranean winter rainfall in phase with African monsoons during the past 1.36 million years. *Nature* 573(7773):256–260. <https://doi.org/10.1038/s41586-019-1529-0>
- Wang P, Tian J, Lourens LJ (2010) Obscuring of long eccentricity cyclicity in Pleistocene oceanic carbon isotope records. *Earth Planet Sci Lett* 290(3):319–330. <https://doi.org/10.1016/j.epsl.2009.12.028>
- Zachos J, Pagani M, Sloan L et al (2001) Trends, rhythms, and aberrations in global climate 65 Ma to present. *Science* 292(5517):686–693. <https://doi.org/10.1126/science.1059412>

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